Seat No : ________

[140]

No. of Printed Pages: 02

SARDAR PATEL UNIVERSITY
BCA Semester-4 Examination – April-2023
US04CBCA27 || OPERATING SYSTEMS

Date : 11-04--2023
Time : 02.00 p.m. – 05.00 p.m. **Total : 70 Marks**

Q-1 Select the appropriate option for the following questions: **10**

1) Compiler is an example of which components of Operating system?
   A. Hardware C. Application Program
   B. System Program D. None
2) A Program in execution is known as _______.
   A. Process C. Commands
   B. Files D. None of these
3) Valid-Invalid bit scheme in page table is used in ________
   A. Fragmentation C. Demand paging
   B. Compaction D. None of these
4) A Lazy Swapper is also known as ________.
   A. Pager C. Pages
   B. Frames D. None of these
5) If resources are not available at requested time then process enters in _____state.
   A. request C. Queue
   B. Wait D. None
6) Each process has a segment of code called _________
   A. important Section C. Mutual Section
   B. Critical Section D. None
7) _______ command is use to display help of LINUX commands.
   A. helpto C. rm
   B. mv D. man
8) _________ command is use to display list of all users currently login on a server.
   A. grep C. search
   B. find D. None
9) Time-sharing system also known as _________.
   A. Real-time C. Multi-user
   B. Multi-tasking D. None of these
10) ________page replacement algorithm will swap a page that will not be used for longer period of time.
   A. FIFO C. LRU
   B. Optimal D. None of these

Q-2 Give short answers of the following questions: (Attempt TEN out of TWELVE) **20**

1) Define Operating system
2) Define Long term scheduler.
3) Draw the diagram of PCB.
4) Explain First-fit memory allocation techniques.
5) What is Belady's Anomaly?

(P.T.O.)

Seat No : ________ No. of Printed Pages: ________

6) Calculate Page faults using Optimal page replacement algorithm for following reference string: (Number of Frames = 3)
Reference string = 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1

7) Explain any two necessary conditions of Critical section.

8) When Race conditions arise?

9) Explain LINUX History in short.

10) Explain ls –l command.

11) Explain rm command with all its attributes

12) Explain mv command with all its attributes

Q-3	**A**	Explain Time Sharing Operating System.	**06**
	B	Explain Layered approach.	**04**
		OR	
Q-3	**A**	Explain FCFS scheduling algorithm in brief.	**06**
	B	Which are the functions performed by Operating System?	**04**
Q-4	**A**	Explain FIFO Page replacement algorithm in detail.	**05**
	B	What is Fragmentation? List different types of Fragmentation. Explain in details.	**05**
		OR	
Q-4	**A**	Explain Swapping of two processes with suitable example.	**05**
	B	Explain Page Fault in Demand Paging. Write step for handing Page Fault.	**05**
Q-5	**A**	What is LINUX? Explain basic features of LINUX Operating System.	**05**
	B	What do you mean by Deadlock? Explain all necessary conditions for occurrence of deadlock.	**05**
		OR	
Q-5	**A**	What is Process Synchronization? Explain.	**05**
	B	Explain Critical-section problem in details. Explain algorithm 1 for solving critical section problem for two-process.	**05**
Q-6		Explain grep command and chmod command in detail.	**[10]**
		OR	
Q-6		Explain all control structures in details.	**[10]**

——X——